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Teaching machine learning at university

A 14-week, 3-credit module in machine learning for third-year computer science or engineering students. Prerequisites: linear algebra, multivariable calculus, probability and Python. The four Darsi lectures anchor weeks 2, 5, 8 and 11.

Learning outcomes

1. Formulate learning as empirical risk minimisation and derive gradient updates.
2. Implement and diagnose gradient descent and SGD from loss curves.
3. Derive and implement backpropagation and verify gradients numerically.
4. Explain and implement scaled dot-product and multi-head attention.
5. Evaluate models without leakage and document them in a model card.

14-week syllabus

Week	Topic	In class and out
1	Supervised learning and loss functions	Lecture; Python refresher lab
2	Empirical risk and gradient descent	Darsi lecture 1; steps by hand
3	SGD, momentum and Adam	Optimisation lab; problem set 1
4	Linear and logistic models	Lecture; quiz 1
5	Neural networks and backpropagation	Darsi lecture 2; three-node worked example
6	Autodiff and gradient checking	Micrograd assignment out
7	Training deep networks	Initialisation, normalisation, residuals
8	Attention	Darsi lecture 3; one head by hand
9	Transformers	Attention-from-scratch lab
10	Language models and fine-tuning	Seminar; problem set 2
11	Evaluation, leakage and model cards	Darsi lecture 4; project proposal due
12	Fairness and disaggregated evaluation	Audit workshop
13	Project clinic	Draft model cards; peer review
14	Final project defences	Ten-minute oral defences

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Lecture slide outline

Lecture 1: Learning as optimisation

- Hook: the nan loss
- Empirical risk $\hat{R}(\theta)$
- $\theta \leftarrow \theta - \eta \nabla \hat{R}$ by hand on θ^2
- Learning-rate regimes
- Mini-batch SGD

Lecture 2: Backpropagation

- Computational graphs
- The chain rule at a node
- Three-node worked example
- Vanishing gradients
- Finite-difference checks

Lecture 3: Attention and transformers

- Hook: what does “it” refer to?
- Q, K, V and $\text{softmax}(QK^T/\sqrt{d})V$
- Why $\sqrt{d}$
- Heads, residuals, positions, masks
- $O(n^2)$ cost

Lecture 4: Evaluation and assessment

- Train, validation, test
- Selection bias and leakage
- Precision, recall, calibration
- Model cards
- Assessment with LLMs allowed

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Problem set

1. For $L(\theta) = \theta^2$, show that gradient descent converges if and only if $0 < \eta < 1$.

2. Derive $\partial L/\partial w$ for logistic loss $L = -y \log \sigma(wx) - (1 - y) \log(1 - \sigma(wx))$.

3. Backpropagate through $f = (a + b)c$ with $a = 1$, $b = 2$, $c = 3$. Give $\partial f/\partial a$, $\partial f/\partial b$ and $\partial f/\partial c$.

4. If q and k have independent entries with mean 0 and variance 1 in d dimensions, find $\mathrm{Var}(q \cdot k)$ and explain the $\sqrt{d}$ scaling.

5. A classifier on 10,000 cases with 100 positives finds 60 of them and raises 140 false alarms. Give precision and recall.

6. A team reports test accuracy after trying 40 configurations on the test set. Explain the bias and propose a fix.

Solutions

1. θ_{t+1} = (1 - 2η)θ_t, so θ_t = (1 - 2η)^t θ_0, which tends to 0 iff |1 - 2η| < 1, i.e. 0 < η < 1.

2. Using $\sigma' = \sigma(1 - \sigma)$, $\partial L/\partial w = (\sigma(wx) - y)x$.

3. $\partial f/\partial a = \partial f/\partial b = c = 3$; $\partial f/\partial c = a + b = 3$.

4. $\mathrm{Var}(q \cdot k) = \Sigma \mathrm{Var}(q_i k_i) = d$. Dividing by $\sqrt{d}$ gives variance 1, keeping softmax out of saturation.

5. Precision = 60/200 = 30%; recall = 60/100 = 60%.

6. The maximum of 40 noisy estimates is biased upwards. Select on validation or CV, then evaluate once on an untouched test set.

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Lab handout: Attention from scratch

Goal

Implement attention, verify it, and judge what attention maps can explain.

Steps

1. Implement softmax($QK^T/\sqrt{d}$)V in NumPy for one head.
2. Add multi-head attention and a causal mask.
3. Check your output against a library implementation on random inputs (tolerance 1e-6).
4. Remove the $\sqrt{d}$ factor with d = 512 and record the entropy of the attention rows.
5. Visualise attention for a pronoun sentence with a pretrained model and write a one-page critique.

Hand in

The notebook (reruns with one command), the tests and the critique.

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Assessment rubric

Criterion	Excellent	Good	Developing	Not yet
Correctness	All tests pass; edge cases handled	All tests pass	Most tests pass	Core function incorrect
Mathematical reasoning	Derivations complete and connected to results	Derivations complete	Partial derivations	No derivation
Evaluation	Leak-free, disaggregated, with uncertainty	Leak-free and appropriate metrics	Some leakage or weak metrics	Invalid evaluation
Oral defence	Explains every choice and its alternatives	Explains most choices	Explains some choices	Cannot explain own work

Reading list

Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.
Chapters 5, 6, 8: learning, backprop, optimisation.

Rumelhart, D. E., Hinton, G. E., & Williams, R. J. (1986). Learning representations by back-propagating errors. Nature, 323, 533–536.
The backpropagation paper.

Vaswani, A., et al. (2017). Attention is all you need. Advances in Neural Information Processing Systems 30.
The transformer.

Bishop, C. M. (2006). Pattern Recognition and Machine Learning. Springer.
Probabilistic foundations.

Russell, S., & Norvig, P. (2020). Artificial Intelligence: A Modern Approach (4th ed.). Pearson.
Broad context for the course.

Mitchell, M., et al. (2019). Model cards for model reporting. Proceedings of FAT* 2019, 220–229.
Documenting models.

Kingma, D. P., & Ba, J. (2015). Adam: A method for stochastic optimization. ICLR 2015.
The Adam optimiser.

Sweller, J. (1988). Cognitive load during problem solving: effects on learning. Cognitive Science, 12(2), 257–285.
Why worked examples help novices.

Freeman, S., et al. (2014). Active learning increases student performance in science, engineering, and mathematics. PNAS, 111(23), 8410–8415.
Evidence for active learning.

Biggs, J., & Tang, C. (2011). Teaching for Quality Learning at University (4th ed.). Open University Press.
Constructive alignment.